

An Internship Report

Of

WEATHER DATA ANALYSIS AND PREDICTION

Submitted

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Roll Number of Student:- 25SCS1003001209

Name of Student: ROHIT KURUP

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CANDIDATE'S DECLARATION

I, ROHIT KURUP, do hereby declare that the internship report titled "WEATHER DATA ANALYSIS AND PREDICTION" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ROHIT KURUP

Roll No.: 25SCS1003001209

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Signature:

Student Name: ROHIT KURUP

Roll No.: 25SCS1003001209

ABSTRACT

Weather affects nearly every sector of human activity, from agriculture and aviation to disaster management and daily planning. This report presents a detailed study of a Weather Data Analysis and Prediction system that analyses historical weather records and predicts future temperature trends using time-series and regression-based techniques.

Weather data is generated continuously from ground stations, satellites, radar networks and ocean buoys, producing large volumes of time-stamped observations such as temperature, humidity, rainfall, wind speed and atmospheric pressure. Making sense of this data requires systematic collection, cleaning and exploratory analysis before any predictive model can be built.

The report describes a modular architecture that includes a data-ingestion and ETL layer, a preprocessing and feature-engineering stage, a model-training and evaluation layer covering linear and polynomial regression, ARIMA and LSTM-based approaches, and a visualization dashboard for presenting forecasts. Core functional modules such as historical data exploration, trend and seasonality analysis, model training, accuracy evaluation and future-temperature forecasting are examined in depth.

Challenges including missing and noisy sensor readings, the non-stationary nature of climate data, and the trade-off between model interpretability and accuracy are analysed, along with opportunities arising from IoT-based sensor networks, hybrid statistical-plus-deep-learning models and localized micro-forecasting. The study concludes with recommendations for building reliable, extensible weather analysis and prediction systems using time-series data and basic regression techniques.

Keywords: Weather Data Analysis, Time Series Forecasting, Regression Techniques, ARIMA, Temperature Prediction, Data Preprocessing, System Design.

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1. INTRODUCTION

1.1 Background

Weather influences agriculture, transportation, energy demand, public health and disaster preparedness. Historically, weather forecasting relied on manual observation and simple statistical averages. The growth of digital weather stations, satellites and open weather APIs has made large volumes of historical and real-time meteorological data available, creating an opportunity to apply data analysis and machine-learning techniques to understand climate patterns and predict future conditions.

A Weather Data Analysis and Prediction system processes historical readings of temperature, humidity, rainfall and other parameters to identify trends, seasonal cycles and anomalies, and then applies time-series or regression-based models to forecast future values — most commonly future temperature trends.

1.2 Problem Statement

Raw weather data is noisy, irregularly sampled and often contains missing values due to sensor faults or transmission gaps. Manually inspecting years of daily or hourly readings to spot trends is impractical. A systematic analysis-and-prediction pipeline is needed to clean the data, visualise historical patterns, and generate reliable short- and medium-term temperature forecasts that can support planning decisions in agriculture, energy and disaster management.

1.3 Scope of the Project

This report covers the conceptual design and detailed analysis of a Weather Data Analysis and Prediction system. The scope includes a review of weather-data sources and time-series concepts, functional and non-functional requirements, system architecture, core modules for analysis and prediction, technology choices, implementation methodology, challenges and future trends. The primary predictive focus is on temperature trend forecasting using time-series and basic regression techniques; the emphasis is on sound system design and methodology rather than a full production deployment.

1.4 Organisation of the Report

Chapter 2 reviews literature and presents an overview of weather data sources and volumes. Chapter 3 lists the objectives. Chapters 4–7 describe requirements, architecture, features and technology stack. Chapters 8–11 discuss methodology, challenges, SWOT and future trends. Chapter 12 concludes the study.

2. LITERATURE REVIEW AND DATA OVERVIEW

2.1 Growth of Weather Data

Meteorological data collection has expanded dramatically with the deployment of automated weather stations, Doppler radar networks, geostationary and polar-orbiting satellites, and ocean buoys. Global meteorological agencies and open-data initiatives now publish decades of historical observations, enabling large-scale time-series analysis that was previously impractical.

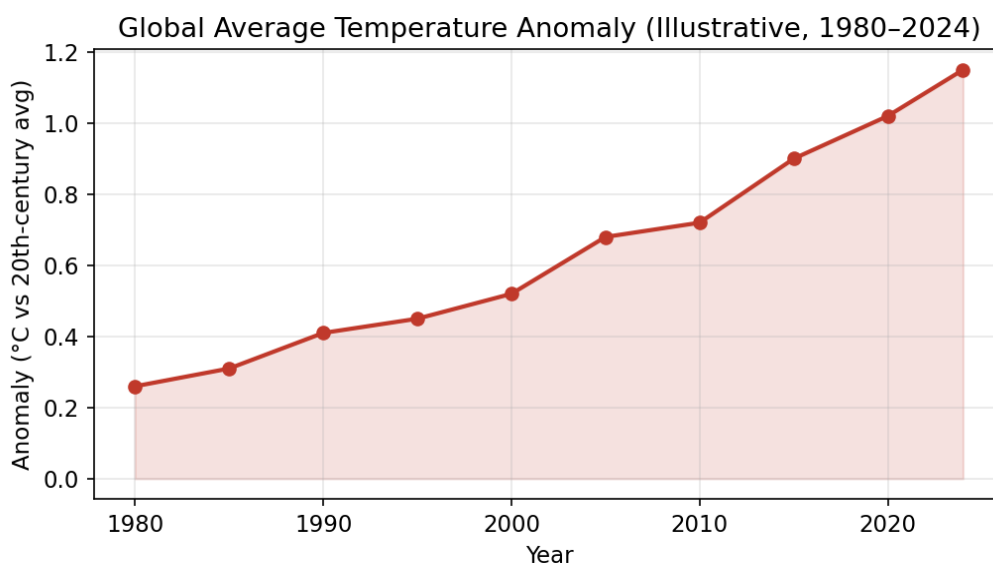


Figure 2.1: Global Average Temperature Anomaly (Illustrative, 1980–2024)

Long-term temperature records collected from ground stations and satellites show a clear warming trend over recent decades. Such long-run historical series form the foundation for trend analysis and are typically the first dataset explored in a weather-prediction project.

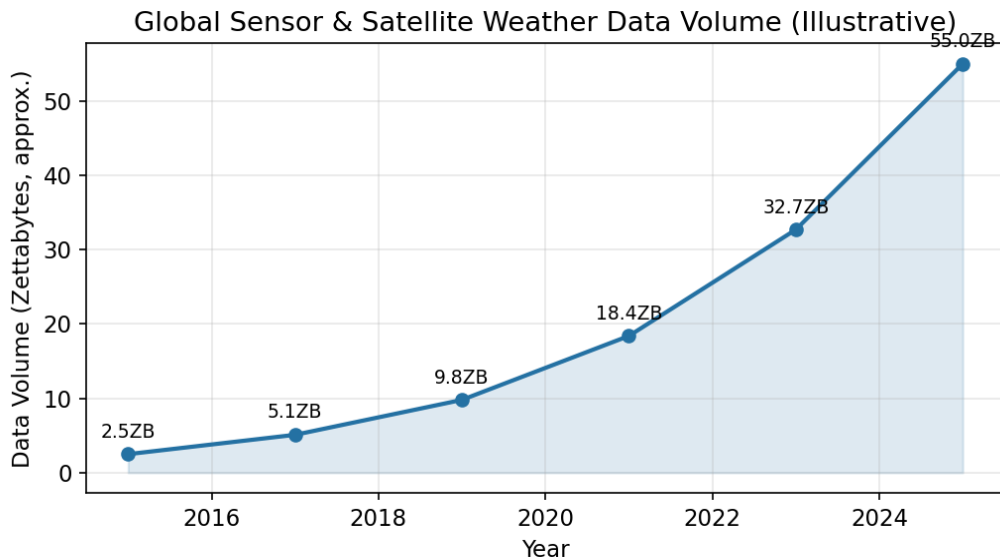


Figure 2.2: Illustrative Growth in Global Weather Data Volume

The volume of weather-related data generated globally has grown steadily as sensor networks and satellite coverage have expanded, increasing both the opportunity and the computational burden of large-scale weather analysis.

2.2 Typical Weather Data Sources

Weather datasets used for analysis and prediction projects are commonly drawn from a mix of sources, each with different resolution, coverage and reliability characteristics.

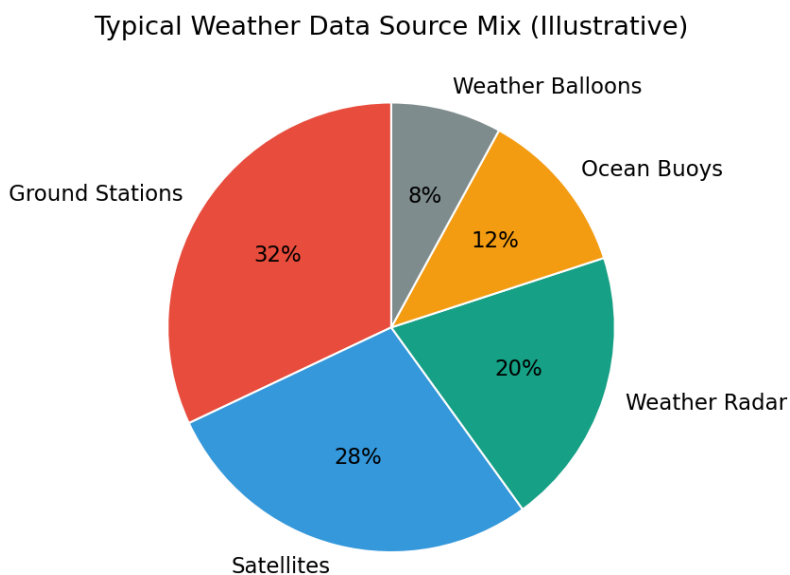


Figure 2.3: Typical Weather Data Source Mix (Illustrative)

- Ground weather stations — direct, high-accuracy point measurements of temperature, humidity, rainfall and pressure.
- Satellites — wide-area coverage of cloud cover, surface temperature and precipitation estimates.
- Weather radar — short-term precipitation intensity and storm tracking.
- Ocean buoys — sea-surface temperature and marine weather parameters relevant to coastal forecasting.
- Weather balloons (radiosondes) — vertical atmospheric profiles used in numerical weather prediction.
- Public APIs and open datasets (e.g. national meteorological department archives, Kaggle weather datasets) — convenient historical CSV/JSON data for analysis projects.

2.3 Time-Series Concepts Relevant to Weather Data

Weather observations are inherently time-series data: values recorded sequentially at regular intervals (hourly, daily, monthly). Three components are typically analysed — the long-term trend (e.g. gradual warming), seasonality (daily and yearly cycles such as summer/winter temperature swings) and residual noise from short-term weather variability. Techniques such as moving averages, decomposition, autocorrelation analysis and stationarity testing (e.g. the Augmented Dickey-Fuller test) are commonly used before model fitting.

2.4 Related Technical Literature

Literature on weather forecasting spans classical statistical methods and modern machine learning. Linear and polynomial regression provide interpretable baselines for trend estimation. ARIMA (AutoRegressive Integrated Moving Average) and its seasonal variant SARIMA are widely used for univariate time-series forecasting. Tree-based ensemble methods such as Random Forest and Gradient Boosting capture non-linear relationships between multiple weather variables. Deep-learning

approaches, particularly Long Short-Term Memory (LSTM) recurrent neural networks, are increasingly applied where long historical sequences and multiple correlated features are available.

3. OBJECTIVES OF THE STUDY

The primary objectives of this project report are as follows:

- To study historical weather data sources, formats and the key characteristics of meteorological time-series data.
- To define clear functional and non-functional requirements for a weather data analysis and prediction system.
- To design a modular system architecture covering data ingestion, preprocessing, model training and visualization.
- To describe core modules including trend analysis, seasonality detection, model training, evaluation and temperature forecasting.
- To compare basic regression techniques and time-series models suited to temperature-trend prediction.
- To recommend an appropriate technology stack and implementation methodology.
- To analyse challenges, perform a SWOT analysis and identify future opportunities and trends.
- To provide practical recommendations for students and practitioners building similar systems.

4. SYSTEM REQUIREMENTS

4.1 Functional Requirements

4.1.1 Data Acquisition

- Import historical weather data from CSV files, public APIs or open datasets.
- Support multiple parameters: temperature, humidity, rainfall, wind speed and atmospheric pressure.
- Allow selection of date range and location/station for analysis.

4.1.2 Data Preprocessing and Exploration

- Detect and handle missing or erroneous readings (interpolation, imputation).
- Resample data to consistent time intervals (daily, monthly, yearly).
- Visualise historical trends, seasonal patterns and anomalies.
- Compute summary statistics (mean, min, max, standard deviation) per period.

4.1.3 Prediction and Modelling

- Train regression and time-series models on historical data to predict future temperature.
- Support model comparison using standard error metrics (RMSE, MAE, R^2).
- Generate short-term and medium-term temperature forecasts with confidence indication.
- Allow retraining as new historical data becomes available.

4.1.4 Visualization and Reporting

- Interactive charts for historical trends and predicted values.
- Downloadable reports/summaries of analysis and forecasts.
- Dashboard view comparing actual vs predicted values.

4.2 Non-Functional Requirements

- Accuracy: Forecast errors should remain within an acceptable range validated against held-out historical data.
- Performance: Data preprocessing and model inference should complete within a few seconds for typical dataset sizes.
- Scalability: Support growing historical datasets spanning multiple decades and multiple locations.
- Reliability: Handle missing or delayed data feeds gracefully without crashing the pipeline.
- Maintainability: Modular pipeline stages (ingestion, cleaning, modelling, visualization) that can be updated independently.
- Usability: Clear visualizations that are interpretable by non-technical stakeholders such as farmers or planners.

5. SYSTEM ARCHITECTURE AND DESIGN

5.1 High-Level Architecture

A Weather Data Analysis and Prediction system is best designed as a modular pipeline that separates data ingestion, preprocessing, model training and presentation. This separation allows each stage to be improved or replaced independently — for example, swapping a regression model for an LSTM network without changing the ingestion or dashboard layers.

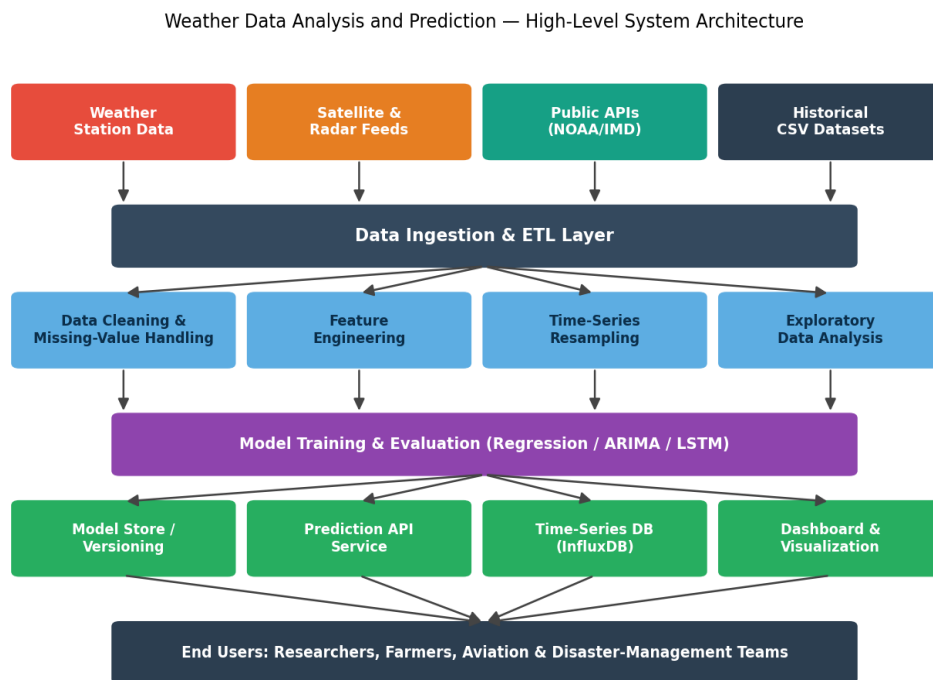


Figure 5.1: High-Level System Architecture of Weather Data Analysis and Prediction System

5.2 Key Components

5.2.1 Data Sources

Historical weather station data, satellite and radar feeds, public meteorological APIs (such as national weather-department portals) and pre-compiled historical CSV datasets feed into the pipeline as raw input.

5.2.2 Data Ingestion and ETL Layer

This layer extracts data from each source, standardises units and timestamps, and loads it into a common time-indexed format ready for cleaning and analysis.

5.2.3 Preprocessing Services

- Data Cleaning & Missing-Value Handling — outlier detection and interpolation of gaps in sensor readings.
- Feature Engineering — deriving features such as day-of-year, rolling averages, lagged temperature values and heating/cooling degree days.

- Time-Series Resampling — converting irregular or high-frequency readings into consistent daily/monthly series.
- Exploratory Data Analysis — trend, seasonality and correlation analysis to guide model choice.

5.2.4 Model Training and Evaluation

Candidate models — linear regression, polynomial regression, ARIMA and, where sufficient data and compute are available, LSTM networks — are trained on historical data and evaluated on a held-out test period using RMSE, MAE and R^2 to select the best-performing approach for the given location and horizon.

5.2.5 Storage and Serving

- Model Store / Versioning — retains trained model artefacts and metadata for reproducibility.
- Prediction API Service — exposes an endpoint that returns forecasted values for a requested date range.
- Time-Series Database (e.g. InfluxDB) — efficiently stores and queries large volumes of timestamped observations.
- Dashboard & Visualization — presents historical trends and forecasts to end users.

5.3 Analysis and Prediction Workflow

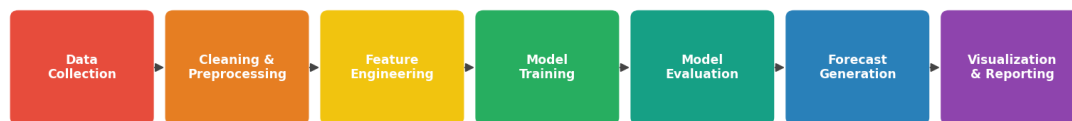


Figure 5.2: End-to-End Weather Data Analysis and Prediction Process Flow

The typical workflow is: historical data is collected from one or more sources → cleaned and preprocessed → relevant features are engineered → a regression or time-series model is trained → the model is evaluated against held-out data → forecasts are generated for the desired horizon → results are visualised and reported to end users.

5.4 Design Principles

- Separation of concerns between data handling, modelling and presentation layers.
- Reproducibility — every trained model is tied to the exact dataset version and parameters used.
- Graceful handling of missing or delayed data without halting the entire pipeline.
- Model transparency — favouring interpretable techniques (regression, ARIMA) alongside more complex models.
- Extensibility — new weather parameters or data sources can be added without redesigning the pipeline.

6. CORE FEATURES AND MODULES

6.1 Historical Data Exploration

The system allows users to browse historical weather records for a chosen location and time range, view raw time-series plots, and compute summary statistics such as monthly averages, extremes and variability. This exploratory step is essential for spotting data-quality issues before modelling begins.

6.2 Trend and Seasonality Analysis

Decomposition techniques separate a temperature series into trend, seasonal and residual components. This helps distinguish a genuine long-term warming or cooling trend from expected seasonal fluctuations, and highlights anomalous years or events.

6.3 Data Cleaning and Preprocessing

Missing readings are imputed using interpolation or seasonal averages; outliers caused by sensor faults are flagged and corrected. Data is resampled to a consistent frequency (daily or monthly) so that models receive uniformly spaced input.

6.4 Feature Engineering

Derived features such as lagged temperature values, rolling means, day-of-year encodings and interaction terms with humidity or rainfall improve the predictive power of regression-based models.

6.5 Regression-Based Trend Modelling

Linear and polynomial regression models fit a mathematical curve to historical temperature data against time, providing an interpretable baseline for long-term trend prediction. These models are simple to train and explain but assume a relatively smooth underlying trend.

6.6 Time-Series Forecasting (ARIMA / LSTM)

ARIMA models capture autocorrelation and short-term dependencies in the temperature series, while LSTM networks can learn more complex, non-linear temporal patterns when sufficient historical data is available. Both approaches are evaluated alongside the simpler regression baselines.

6.7 Model Evaluation and Comparison

Illustrative Model Comparison — Temperature Prediction RMSE (°C, lower is better)

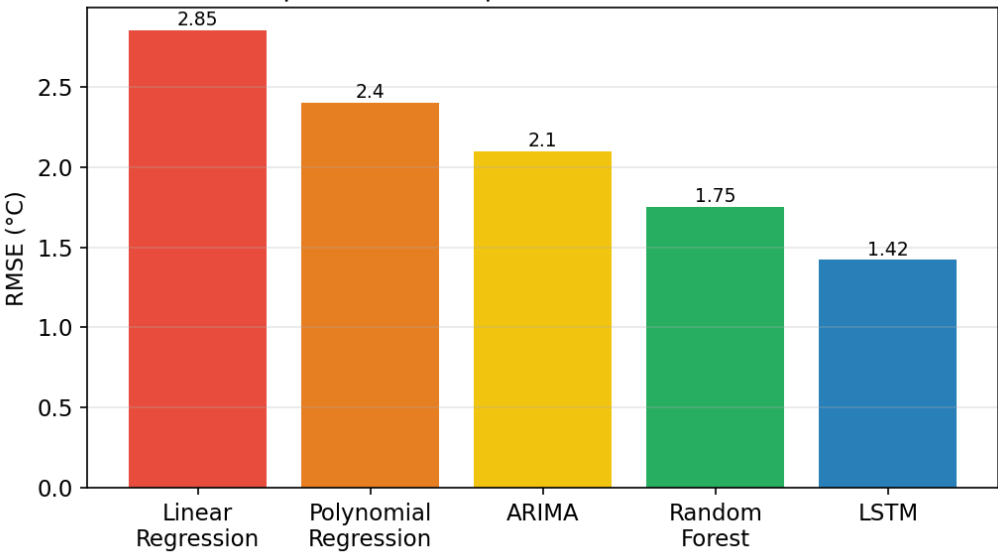


Figure 6.1: Illustrative Model Comparison — Temperature Prediction RMSE

Models are compared using RMSE, MAE and R^2 on a held-out test period. Illustrative results typically show simple regression models providing a fast, interpretable baseline, while ARIMA and LSTM-based models can achieve lower error at the cost of additional complexity and tuning effort.

6.8 Forecast Visualization and Reporting

Predicted temperature trends are plotted alongside historical actuals so that users can visually assess forecast quality. Dashboards summarise upcoming expected conditions and flag periods of higher forecast uncertainty.

7. TECHNOLOGY STACK

7.1 Data Analysis and Modelling

- Language: Python — the standard choice for data analysis and machine learning.
- Data handling: Pandas and NumPy for cleaning, transformation and numerical computation.
- Visualization: Matplotlib and Seaborn for exploratory plots; Plotly for interactive dashboards.
- Regression & ML models: scikit-learn (Linear Regression, Polynomial Regression, Random Forest).
- Time-series modelling: statsmodels (ARIMA/SARIMA); TensorFlow/Keras or PyTorch for LSTM models.

7.2 Data Storage

- CSV/Parquet files for smaller historical datasets and reproducible experiments.
- Time-series database (e.g. InfluxDB or TimescaleDB) for larger-scale, continuously updated deployments.
- Relational database (PostgreSQL/SQLite) for metadata, station information and model run logs.

7.3 Backend and Serving

- API framework: Flask or FastAPI to expose a prediction endpoint.
- Scheduling: cron jobs or Apache Airflow for periodic retraining as new data arrives.

7.4 Frontend and Visualization

- Dashboard: Streamlit or a React-based web app for interactive exploration of trends and forecasts.
- Charting libraries: Plotly.js or Chart.js for interactive frontend visualizations.

7.5 Development and Notebooks

- Jupyter Notebook / Google Colab for exploratory analysis and model prototyping.
- Git for version control of code and experiment tracking.

7.6 Third-Party Data Sources

- Public weather APIs (e.g. national meteorological department portals, OpenWeatherMap-style services).
- Open historical weather datasets (e.g. Kaggle weather datasets, NOAA/IMD archives).

8. IMPLEMENTATION METHODOLOGY

8.1 Iterative Analysis Approach

An iterative, notebook-driven methodology is recommended, where each stage — data collection, cleaning, exploration, modelling and evaluation — is refined in short cycles as insights from earlier steps inform later ones. This mirrors typical data-science workflow practice rather than a rigid waterfall process.

8.2 Development Phases

- Phase 1 — Data collection and understanding: gather historical datasets, review data dictionaries and identify quality issues.
- Phase 2 — Exploratory data analysis: visualise trends, seasonality and correlations between weather variables.
- Phase 3 — Preprocessing and feature engineering: handle missing values, resample series, derive predictive features.
- Phase 4 — Model building: train baseline regression models, then time-series models (ARIMA) and optionally LSTM.
- Phase 5 — Evaluation and forecasting: compare models on held-out data, select the best performer, generate forecasts and build the reporting dashboard.

8.3 Testing Strategy

- Validation split: train/test split respecting time order (no shuffling) to avoid data leakage.
- Cross-validation: time-series cross-validation (rolling-origin) for more robust performance estimates.
- Error analysis: inspecting residuals for remaining trend or seasonality not captured by the model.
- Sanity checks: comparing forecasts against simple baselines such as persistence ("tomorrow = today") or seasonal-naïve forecasts.

8.4 Deployment Considerations

For a student project, deployment can be as simple as a scheduled script that retrains the model monthly and updates a dashboard. In a production setting, blue-green deployment of updated models and monitoring of forecast accuracy over time would be used to catch model drift as climate patterns evolve.

9. CHALLENGES AND RISK ANALYSIS

9.1 Data Challenges

- Missing or inconsistent readings due to sensor faults, maintenance gaps or station relocation.
- Inconsistent units, timestamps or formats when combining multiple data sources.
- Limited historical depth for some locations, restricting the length of usable training data.

9.2 Modelling Challenges

- Non-stationarity — long-term climate trends mean older data may not fully represent future conditions.
- Extreme events (heatwaves, cold snaps) are rare and harder for standard regression models to capture.
- Overfitting when using complex models (e.g. LSTM) on relatively small historical datasets.
- Balancing model interpretability (regression, ARIMA) against raw predictive accuracy (deep learning).

9.3 Operational Risks

- Dependence on third-party APIs that may change formats, rate-limit requests or become unavailable.
- Need for periodic retraining as new data arrives to avoid model drift.
- Communicating forecast uncertainty clearly so users do not over-trust point predictions.

9.4 Mitigation Strategies

Robust preprocessing pipelines with automated data-quality checks, use of ensemble or hybrid models that combine interpretable and complex approaches, regular retraining schedules, and clear visualisation of confidence intervals alongside forecasts help mitigate many of the above risks.

10. SWOT ANALYSIS

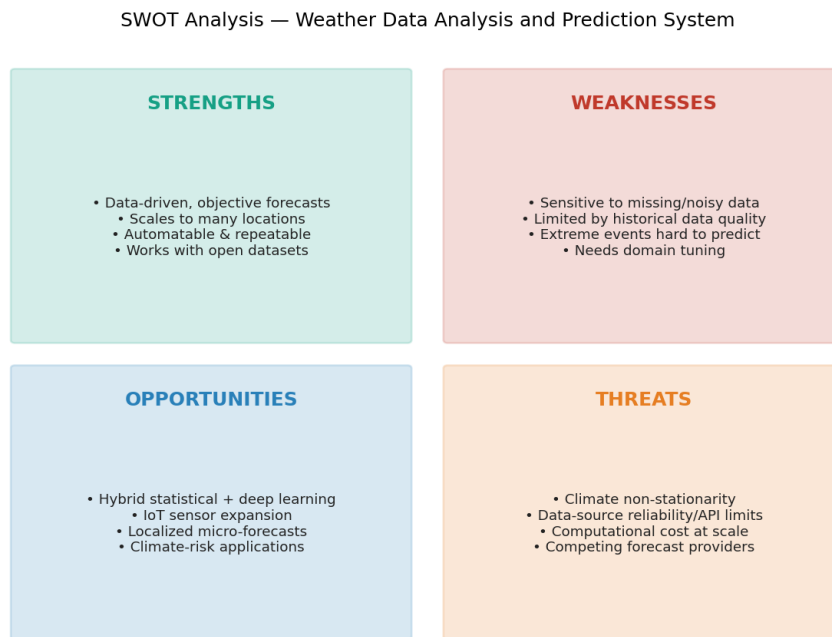


Figure 10.1: SWOT Analysis of Weather Data Analysis and Prediction System

10.1 Strengths

Data-driven forecasts are objective and repeatable, unlike purely manual observation. The approach scales across many locations and time periods once the pipeline is built, and can be automated to run with minimal manual intervention. Open historical datasets make the approach accessible for academic and small-scale projects.

10.2 Weaknesses

Model quality is sensitive to missing or noisy input data. Historical data quality and length vary significantly by location, which can limit forecast reliability. Extreme or rare weather events remain difficult for standard regression and basic time-series models to predict accurately.

10.3 Opportunities

Combining statistical models with deep learning (hybrid approaches), integrating growing IoT sensor networks for finer-grained data, producing hyper-local micro-forecasts, and applying the same techniques to climate-risk assessment represent significant opportunities for extending this project.

10.4 Threats

Climate non-stationarity means patterns learned from historical data may shift over time. Reliance on external APIs introduces availability and format-change risk. Computational cost can grow significantly when scaling to many locations or more complex models, and established commercial forecast providers set a high bar for accuracy.

11. FUTURE TRENDS AND OPPORTUNITIES

11.1 Hybrid Statistical and Deep-Learning Models

Combining interpretable statistical models (regression, ARIMA) with deep-learning architectures (LSTM, Transformer-based time-series models) is an active area, aiming to capture both clear long-term trends and complex short-term dynamics.

11.2 IoT and Sensor Network Expansion

Low-cost IoT weather sensors are making hyper-local data collection increasingly feasible, enabling finer-grained analysis than traditional sparse weather-station networks.

11.3 Localized Micro-Forecasting

As data density increases, forecasting is moving from broad regional predictions toward highly localized, even street-level, temperature and precipitation forecasts relevant to agriculture and urban planning.

11.4 Climate-Risk and Decision-Support Applications

Weather trend analysis increasingly feeds into longer-horizon climate-risk applications — crop planning, energy-demand forecasting, and disaster early-warning systems — extending beyond short-term weather prediction.

11.5 Explainable and Uncertainty-Aware Forecasting

Future systems are expected to place greater emphasis on communicating forecast uncertainty (confidence intervals, probabilistic forecasts) and on explaining which historical patterns drove a given prediction, building user trust in automated forecasts.

12. CONCLUSION AND RECOMMENDATIONS

12.1 Conclusion

A Weather Data Analysis and Prediction system demonstrates how time-series data and basic regression techniques can be applied to a practical, high-impact problem. Careful data cleaning and exploratory analysis are as important as model selection itself, since weather data is inherently noisy and irregularly sampled. Simple regression and ARIMA models provide strong, interpretable baselines, while more advanced approaches such as LSTM can improve accuracy when sufficient historical data is available.

The study confirms that a well-structured pipeline — covering data ingestion, preprocessing, feature engineering, model training, evaluation and visualization — forms a solid technical foundation for reliable temperature-trend forecasting, and that this foundation can be extended to other weather parameters or additional locations with minimal redesign.

12.2 Recommendations

- Start with thorough exploratory data analysis before selecting a modelling approach.
- Always compare simple baselines (persistence, seasonal-naive, linear regression) against more complex models.
- Use time-respecting train/test splits and rolling-origin cross-validation to avoid unrealistic accuracy estimates.
- Invest in robust handling of missing and anomalous sensor data early in the pipeline.
- Communicate forecast uncertainty alongside point predictions rather than presenting a single number.
- Design the pipeline to be extensible to additional weather parameters and locations from the outset.

12.3 Future Scope of Work

Future extensions of this study could include a full working prototype with a live data feed, comparison of additional forecasting algorithms (e.g. Prophet, Transformer-based models), multi-location and multi-parameter forecasting, or integration with a real-time alerting system for extreme-weather events.

13. REFERENCES

1. National meteorological department historical weather data archives (e.g. IMD, NOAA).
2. Kaggle — publicly available historical weather datasets used for time-series analysis projects.
3. Documentation of Pandas, NumPy, scikit-learn, statsmodels and TensorFlow/Keras libraries.
4. Academic and practitioner literature on ARIMA, SARIMA and LSTM-based time-series forecasting.
5. Technical articles on exploratory data analysis and feature engineering for time-series data.
6. Open-source weather API documentation (e.g. national meteorological portals, community weather APIs).
7. Standard references on regression analysis and error metrics (RMSE, MAE, R^2).
8. Publicly available blogs and tutorials describing weather-forecasting pipelines and dashboards.

APPENDIX

A. Glossary of Terms

RMSE — Root Mean Squared Error: a common metric for regression/forecast accuracy, penalising larger errors more heavily.

MAE — Mean Absolute Error: the average magnitude of forecast errors, in the same units as the target variable.

R² — Coefficient of Determination: the proportion of variance in the target explained by the model.

ARIMA — AutoRegressive Integrated Moving Average: a classical statistical model for time-series forecasting.

LSTM — Long Short-Term Memory: a type of recurrent neural network well suited to sequential/time-series data.

Stationarity — a property of a time series whose statistical characteristics (mean, variance) do not change over time; often required or tested for before applying ARIMA-style models.

Seasonality — regular, periodic fluctuations in a time series, such as annual temperature cycles.

B. Sample Non-Functional Targets

Forecast horizon: short-term (1–7 days) and medium-term (1–3 months) temperature predictions.

Target accuracy: RMSE within an acceptable range validated against historical held-out data for the chosen location.

Data refresh frequency: daily ingestion of new observations where a live data feed is available.

Model retraining cadence: monthly or seasonal retraining to incorporate the latest historical data.

C. Declaration

I hereby declare that the project report entitled "Weather Data Analysis and Prediction" is a record of original work done by me under the guidance of the faculty supervisor. The information and data presented are based on publicly available sources and standard data-analysis practices. Any resemblance to proprietary internal systems of commercial weather-forecasting providers is coincidental and used only for academic illustration.

Date: _____

Signature of the Student: _____